

INTELLIGENT LIBRARY ASSISTANCE BOT

A PROJECT REPORT

Submitted by

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*in partial fulfilment for the award of the degree
of*

BACHELOR OF ENGINEERING

in

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

R.M.K. ENGINEERING COLLEGE

(An Autonomous Institution) R.S.M. Nagar, Kavaraipettai-601206



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BONAFIDE CERTIFICATE

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ABSTRACT

Libraries often encounter challenges in quickly locating and retrieving requested books, which leads to increased waiting times and greater manual effort by the staff. To address this problem, an **Automated Library Assistant Robot** is designed to navigate through the library along a **straight, predefined path** while **identifying books using QR code scanning** technology.

The robot moves systematically through the library aisles, **continuously scanning QR codes** that are placed on the racks. Each QR code contains information about the books on that particular shelf. When the robot scans a code that matches the **user's requested book details**, it automatically **stops at the corresponding shelf**. At this point, a **robotic gripper mechanism** is activated to **pick the required book** from the rack and **place it safely in a holding tray**.

Once the book is collected, the robot **retraces its path back to the librarian's desk**, where it delivers the book to the user. This automated process **minimizes human intervention, reduces retrieval time, and enhances the overall efficiency** of library operations. By integrating robotics and automation, this system provides a **smart, reliable, and user-friendly solution** for modern library management.

Keywords: Library Automation, Service Robot, QR Code Scanning, Robotic Gripper, Pick and Place

TABLE OF CONTENTS

CH APT ER NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	
	ABSTRACT	
	LIST OF FIGURES	
	LIST OF ABBREVIATIONS	
1	INTRODUCTION	
	1.1 Introduction	1
	1.2 Scope Of Project	2
2	OBJECTIVE	3
3	EXISTING METHOD	5
4	LITERATURE REVIEW	7
	4.1 Robotic Library Assistant	7
	4.2 Automation in Library Management Using LabVIEW	7
	4.3 QR Code–Based Navigation System for Autonomous Indoor Robots	8
	4.4 IoT-Enabled Smart Library Monitoring and Retrieval System	8
	4.5 Vision-Based Library Robot for Book Identification and Shelf Navigation	9
5	PROPOSED SYSTEM	10
	5.1 General Description	11
	5.2 Block Diagram	12
	5.3 Working Principle	
6	HARDWARE AND SOFTWARE COMPONENTS	
	6.1 HARDWARE COMPONENTS	13
	6.1.1 Raspberry Pi 4	13
	6.1.2 DC Motors	15
	6.1.3 Raspberry Pi Camera Module	16
	6.1.4 L298n Motor Driver Module	17
	6.1.5 IR Sensor Module	18

6.2	SOFTWARE REQUIREMENTS	19
6.2.1	Raspberry Pi OS	19
6.2.2	Python,Open Cv	20
6.2.3	Google Cloud	20
7	FINAL OUTPUT	
7.1	Hardware Setup	21
7.2	Code	21
7.3	OUTPUT	
7.3.1	Software Output	24
8	APPLICATIONS AND FUTURE ENHANCMENT	
8.1	Applications	25
8.2	Future Enhancement	27
9	CONCLUSION	28
10	REFERENCES	29

LIST OF FIGURES

FIGURE NO.	DESCRIPTION	PAGE NO
5.2	BLOCK DIAGRAM	11
6.1.1	RASPBERRY PI 4	13
6.1.2	DC MOTOR	15
6.1.3	RASPBERRY PI CAMERA MODULE	16
6.1.4	L298N MOTOR DRIVER	17
6.1.5	IR SENSOR MODULE	18
6.2.1	RASPBERRY PI OS	19
6.2.2	PYTHON	20
6.2.3	GOOGLE CLOUD	20
7.1	HARDWARE SETUP	22
7.3.1	SOFTWARE OUTPUT	24

LIST OF ABBREVIATIONS

Abbreviation	Expansion
IOT	Internet Of Things
OLED	Organic Light emitting Diode
OpenCV	Open-source Computer Vision library
AI	Artificial Intelligence
ML	Machine Learning
IR	Infrared (sensor)
QR	Quick Response
DC	Direct Current (motors)

CHAPTER - 1

INTRODUCTION

1.1 GENERAL DESCRIPTION

Libraries serve as vital information hubs, storing, organizing, and managing vast collections of books, journals, and academic resources. In large libraries, the manual process of locating and retrieving books can be time-consuming, labor-intensive, and prone to human errors. Librarians often spend considerable effort searching for misplaced books, navigating long aisles, and handling multiple user requests simultaneously. These challenges lead to slower service, reduced operational efficiency, and increased workload on library staff. To overcome these limitations, the Intelligent Library Assistance Bot has been developed as an automated solution to streamline book retrieval and enhance overall library functioning.

The proposed robotic system employs QR code–based navigation and camera-assisted identification to autonomously locate, retrieve, and deliver the requested book. When a user submits a book request at the librarian’s desk, the robot activates and follows a predefined route marked with QR codes placed on the floor. These QR markers guide the robot through Level-1 scanning for aisle detection and Level-2 scanning for identifying the precise shelf and row where the book is located. Once the position is confirmed, the robot uses a robotic arm equipped with a gripper mechanism to securely pick up the book.

Powered by a Raspberry Pi, integrated sensors, and real-time image processing, the system ensures accurate movement, precise book identification, and safe handling. By automating search and retrieval, the Intelligent Library Assistance Bot reduces human effort, minimizes errors, and significantly speeds up library services. This innovation supports smart campus development by introducing robotics and intelligent automation into the traditional library environment, ultimately improving user experience and operational efficiency.

1.2 SCOPE OF PROJECT

The **Intelligent Library Assistance Bot** is designed to operate in **educational, research, and corporate libraries** to automate the process of book retrieval and delivery. The system aims to improve the efficiency of library operations by reducing manual work and ensuring quick, accurate, and reliable book access for users. The project's scope covers both hardware and software development, integrating robotics, embedded systems, and cloud-based data management.

- **QR Code–Based Navigation System:**

QR codes are placed along the library floor and racks to guide the robot through predefined paths. These act as navigation markers, helping the robot determine its position, direction, and movement efficiently without human assistance.

- **Scanning and Identification Modules:**

The robot uses camera modules and sensors for two-level scanning.

- *Level 1:* Identifies the correct aisle based on QR codes.
- *Level 2:* Locates the exact rack and row of the requested book.

- **Robotic Arm with Gripper Mechanism:**

A servo-controlled robotic arm with a flexible gripper securely handles books of various sizes. It applies optimal pressure to pick and place books smoothly without causing damage.

- **Integration of Raspberry Pi and ESP32 Controllers:**

The Raspberry Pi serves as the main processor for navigation, image processing, and decision-making. The ESP32 manages motors, sensors, and wireless communication.

- **Interactive User Interface:**

A simple interface allows users or librarians to enter book details like title or ID. The system then initiates retrieval automatically while displaying real-time updates on location, progress, and delivery status.

- **Cloud Connectivity:**

Integration with Google Sheets, Firebase, or AppSheet enables real-time data storage, tracking, and centralized record management for improved efficiency.

- **Smart Library Infrastructure:**

The system promotes digital transformation by integrating IoT and automation, with potential upgrades like voice commands, AI-based navigation, and automated sorting for future smart libraries.

CHAPTER-2

2 . OBJECTIVE

The **primary objective** of the Intelligent Library Assistance Bot project is to **automate the process of locating, retrieving, and delivering books** within a library environment using **robotics, embedded systems, and QR code–based navigation**. The system is designed to enhance the **efficiency and accuracy** of library operations, **reduce manual intervention**, and **improve the overall user experience** in accessing library resources.

1. Enhanced Efficiency :

One of the key objectives is to **significantly reduce the time and manual effort** required for book retrieval and delivery. The robot can autonomously navigate through library aisles, identify book locations, and fetch the requested items without human guidance. By operating continuously and autonomously, it eliminates the delays caused by manual searching, thereby **increasing the speed and reliability of library operations**. This level of automation ensures smooth workflow even during peak hours or in large-scale library setups.

2. Improved Accuracy :

The system ensures **precise book identification and location tracking** through the use of **QR code scanning technology**. Each rack and row in the library is labeled with a unique QR code, enabling the robot to identify the exact position of each book. Automated database matching allows the robot to verify the correct book before retrieval, ensuring **error-free selection**. This level of accuracy minimizes human mistakes and improves overall reliability in the library's inventory management.

3. Optimized Resource Utilization :

Another important goal is to **maximize the efficiency of human resources**. By taking over repetitive tasks such as locating and delivering books, the Intelligent Library Assistance Bot **frees librarians from manual labor**, allowing them to focus on more meaningful tasks like catalog

maintenance, user assistance, and administrative functions. This optimization not only improves staff productivity but also leads to **better service quality and user engagement**.

4. User Satisfaction :

The project is designed with the end-user in mind. By ensuring **fast, reliable, and automated book delivery**, the system provides a **convenient and seamless experience** for library users. The intelligent nature of the robot demonstrates innovation in service and enhances the reputation of the library as a technologically advanced facility. Users benefit from **shorter waiting times, accurate book retrieval**, and an **interactive interface**, all contributing to higher satisfaction and trust in library services.

5. Cost-Effective Automation:

A major objective of this project is to develop a **low-cost and scalable automation system** using **readily available and affordable components** such as **QR codes, Raspberry Pi, and sensors**. Compared to expensive technologies like **RFID systems** or **Automated Storage and Retrieval Systems (ASRS)**, the proposed design offers a **budget-friendly alternative** while maintaining high functionality and reliability. This makes it a practical solution for educational institutions, small libraries, and research centers with limited budgets.

Overall, the Intelligent Library Assistance Bot is designed to bring **automation, precision, and convenience** to library management. By meeting these objectives, the project aims to revolutionize the traditional library model into a **smart, efficient, and user-centric environment**, promoting the integration of robotics and IoT in everyday information systems.

CHAPTER-3

3. EXISTING METHOD

Traditional library management systems mainly depend on manual or semi-automated operations. In most libraries, both users and librarians are required to physically walk through the aisles, search the shelves, and pick the required books by hand. Although such systems have been used effectively for many years, they are time-consuming and involve continuous human effort. Manual handling also increases the chances of misplaced books and delays during busy hours.

In conventional libraries, even though barcodes or RFID tags are used to track and maintain records, they do not completely remove human involvement. The barcode or RFID tag helps identify the book and record its issue or return, but the user or librarian must still locate and collect the book physically from the shelf. This partial automation helps in maintaining accurate data but fails to make the system fully autonomous.

To enhance efficiency, modern libraries have started adopting technologies such as **Online Public Access Catalogs (OPAC)** and **RFID-based management systems**. OPAC systems allow users to search for books digitally and find information like the book's section, rack, and shelf number. Similarly, RFID tagging helps in quickly scanning and identifying books without the need for line-of-sight scanning like barcodes. However, in both systems, the physical process of retrieving the book is still performed manually. The user must still go to the exact location and take the book from the shelf, meaning complete automation is not achieved.

In large-scale industrial environments, some organizations use **Automated Storage and Retrieval Systems (ASRS)**. These systems employ robotic cranes, conveyors, or mechanical arms to automatically pick and place items. They are extremely efficient in terms of accuracy and speed. However, the main drawback of these systems is their **high cost, complex setup, and large space requirement**. This makes them unsuitable for small or medium-sized libraries, especially in educational institutions where available space and funding are limited. The installation of such systems requires specific infrastructure and heavy machinery, which cannot be easily adapted to existing library layouts.

In recent years, researchers have explored alternative solutions using **QR code-based navigation** and **computer vision technologies**. These systems allow robots to detect and follow paths using QR codes placed on the floor or shelves. Some prototypes have successfully demonstrated basic navigation and localization capabilities, where the robot can move autonomously within the library and identify shelves through visual markers. However, most of these models focus only on navigation and path detection, rather than on the full process of retrieving and delivering books. Therefore, they only achieve partial automation and still rely on human assistance for handling the books.

Despite continuous development in library technology, several **limitations** remain in existing systems:

- **High manual involvement:** Even advanced systems require humans to find and retrieve books manually.
- **Time consumption:** Users spend extra time searching shelves, especially during peak hours.
- **Limited automation:** Barcode and RFID systems mainly support identification and tracking but cannot perform book retrieval or placement.
- **High cost of automation:** Fully automated ASRS solutions are too expensive and not practical for libraries with smaller budgets.
- **Lack of flexibility:** Many automated systems are designed for warehouse environments, not for open-access libraries where layout and usage patterns frequently change.
- **Human error:** Manual handling may lead to misplaced or incorrectly recorded books, affecting accuracy and user satisfaction.

In conclusion, existing systems like **RFID**, **OPAC**, and **ASRS** have certainly improved certain aspects of library management, such as cataloging and tracking. However, they do not offer a **complete, cost-effective, and intelligent automation solution** suitable for everyday library operations. The dependency on manual retrieval still remains a major limitation.

CHAPTER-4

4. LITERATURE REVIEW

4.1 . Robotic Library Assistants

AUTHORS: S. P. Animireddy, K. L. Prabha Singh, Neha, and V. Natarajan (2018)

DESCRIPTION:

This research focused on the design and development of an **autonomous robot** for assisting in book transportation within a library environment. The robot employed **infrared (IR) sensors** and an **Arduino microcontroller** to follow a predefined line path between bookshelves and the librarian's desk. The study highlighted the effectiveness of using a line-following robot in structured environments for reducing manual effort and improving retrieval efficiency. The authors demonstrated that such a robotic system could effectively transport books upon request, reducing librarian workload and serving as a foundation for automation in academic libraries. However, the major limitations included **rigid navigation dependency** on black-line tracks, **lack of flexibility** in changing library layouts, and **absence of real-time data connectivity**. The system also lacked advanced computer vision capabilities to identify books automatically. This paper laid the groundwork for integrating robotics into library automation but revealed that **future systems must include adaptable navigation, vision-based recognition, and cloud connectivity** for improved performance and scalability.

4.2 Automation in Library Management Using LabVIEW

AUTHORS: A. Gade and Y. Angal (2018)

DESCRIPTION:

This study introduced an **RFID-based automated library system** designed to streamline book issuing, returning, and catalog management processes. Each book was tagged with an RFID chip, and RFID readers were strategically installed across the library. The system, developed using **LabVIEW software**, automatically recorded every transaction, updating a central database in real-time.

The proposed system significantly improved efficiency in record keeping, minimized human errors in book tracking, and enhanced the overall operational flow. However, the model's limitation was its **inability to physically retrieve or deliver books**. While it automated data management effectively, it still relied on manual effort for searching and fetching the requested books from shelves. Furthermore, the **high implementation cost** of RFID tags and readers made it less practical for smaller institutions. Nonetheless, this work demonstrated the advantages of digital automation and emphasized the potential for **integrating hardware and software systems** to achieve full library automation.

4.3 QR Code–Based Navigation System for Autonomous Indoor Robots

AUTHORS: S. Raut, S. A. Patil, and V. R. Shinde (2021)

DESCRIPTION:

This paper proposed a **low-cost indoor navigation system** utilizing QR codes as positional markers for mobile robots. The robot used an onboard camera to capture and decode QR codes placed on the floor, which contained coordinate data (x, y) and direction information. The technique allowed the robot to identify its precise location, maintain orientation, and move autonomously along a predefined path.

The authors demonstrated that QR code navigation offered a **cost-effective and highly accurate** alternative to expensive sensors like LiDAR or GPS, especially in controlled environments such as libraries or warehouses. Experimental results showed consistent performance even under varying lighting conditions.

This research forms a direct foundation for the proposed **Intelligent Library Assistance Bot**, which adapts the same concept of QR-based localization and integrates it with robotic gripping and cloud communication. The study validates that QR technology can effectively support **autonomous indoor movement, precision navigation, and real-time positioning** in library automation systems.

4.4 IoT-Enabled Smart Library Monitoring and Retrieval System

AUTHORS: M. Ali, F. Rahman, and D. Joseph (2022)

DESCRIPTION:

This paper presented an IoT-based smart library system that integrated environmental sensing, book tracking, and automated retrieval assistance. The system used ESP32 microcontrollers connected to a network of RFID readers, temperature sensors, and weight-based shelf sensors to monitor book presence and shelf conditions. A cloud dashboard was developed using Google Firebase to store book locations, user requests, and system health data.

Although the system did not include a mobile robot for physical retrieval, it provided advanced digital automation by allowing users to send book requests through a web interface. The system automatically suggested optimal paths for manual retrieval and alerted librarians when books were misplaced. The major limitations were the absence of autonomous navigation and the need for manual handling of requested books. Nonetheless, this research demonstrated the growing importance of IoT and cloud connectivity in building scalable, efficient, and partially automated library ecosystems. It also provided a strong foundation for integrating robotics with IoT to achieve full library automation in future implementations.

4.5 Vision-Based Library Robot for Book Identification and Shelf Navigation

AUTHORS: R. Karthikeyan, P. Harshitha, and A. Manoj (2020)

DESCRIPTION:

This study introduced a vision-based autonomous robot designed to assist in book identification and locating specific shelves within a library. The robot used a Raspberry Pi equipped with an HD camera to capture real-time images of bookshelves. Image processing techniques such as edge detection, template matching, and OCR (Optical Character Recognition) were implemented to read book titles and classify them according to the library's indexing system. The navigation system relied on ultrasonic sensors for obstacle detection and wheel encoders for distance measurement.

The research demonstrated that incorporating computer vision significantly improved accuracy in identifying books without requiring RFID or barcode tagging. However, the system faced limitations such as dependency on consistent lighting and challenges in reading damaged or poorly printed book spines. Additionally, the robot did not include cloud-based synchronization, restricting remote monitoring and advanced request handling. Despite these limitations, this work highlighted the potential of integrating low-cost vision techniques with robotics to automate book searching and categorization inside academic libraries.

CHAPTER - 5

PROPOSED SYSTEM

5.1 GENERAL DESCRIPTION

The **Intelligent Library Assistance Bot (ILAB)** is a fully autonomous robotic system designed to automate the **book retrieval and delivery process** in libraries. The system combines **QR code navigation, computer vision, robotic gripping, and cloud integration** to create a self-operating mechanism capable of identifying, fetching, and delivering requested books without human assistance.

When a user requests a book through the **AppSheet-based user interface** or a library terminal, the system searches the **cloud database** to locate the book's rack and shelf position. Based on this data, the robot initiates movement from its docking station (librarian's desk) and follows a series of **floor-mounted QR codes** that guide its navigation route throughout the library.

A **downward-facing camera** continuously captures these QR codes, and the decoded information helps the robot determine its position, direction, and next movement. Once it reaches the correct aisle, a **secondary front-facing camera** performs fine-level scanning to identify the specific shelf and book location using QR or label data.

The **robotic arm**, equipped with a **servo-controlled gripper**, extends toward the identified position and carefully picks the book. The system ensures precise handling using motor control algorithms and position sensors, preventing any physical damage to books. After successful gripping, the robot returns to the librarian's desk, where it gently places the book in a delivery tray.

Throughout this operation, the **Raspberry Pi 4** serves as the main controller, coordinating between various modules such as cameras, motor drivers, and cloud APIs. It communicates with an **ESP32 microcontroller**, which controls the DC motors and IR sensors responsible for movement, obstacle detection, and speed regulation.

Additionally, the entire retrieval process is **logged in real-time** on a cloud-based Google Sheet via AppSheet integration, allowing the librarian to track every request, completion time, and delivery status. The system also displays current operation messages on an **OLED display**, providing on-device feedback such as “Scanning Path,” “Book Found,” or “Returning to Desk.”

The combination of **hardware intelligence (robotics)** and **software intelligence (cloud and vision algorithms)** makes ILAB a highly adaptable, scalable, and cost-effective solution suitable for educational institutions, research centers, and public libraries.

5.2 BLOCK DIAGRAM:

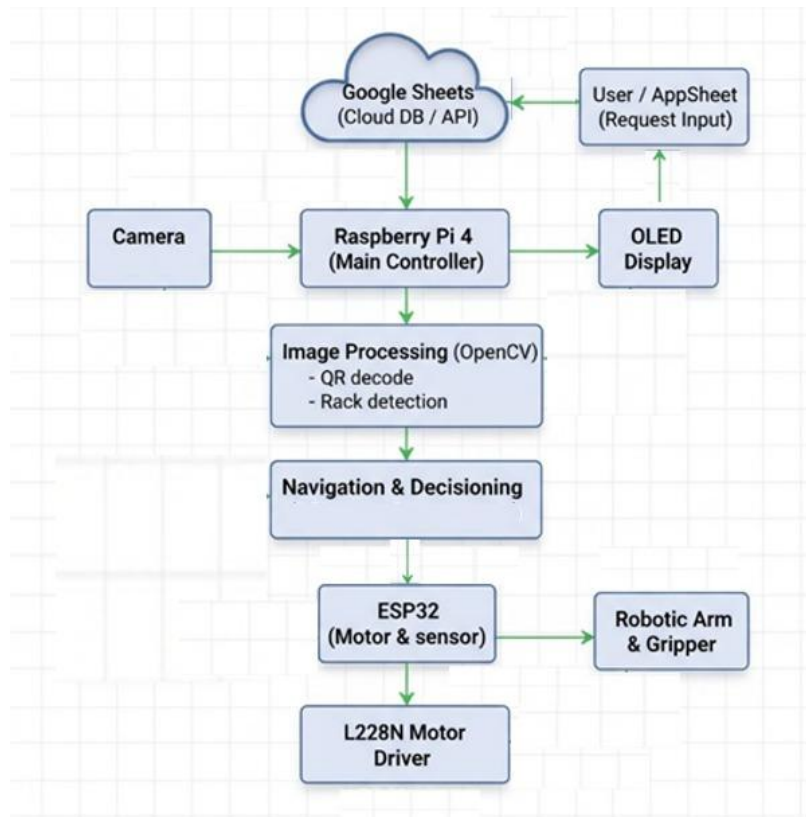


Figure 5.2 : Block Diagram for Intelligent Library Assistance Bot

5.3 WORKING PRINCIPLE:

1. Book Request and Identification

- The user initiates a request by entering the book's title or ID through the AppSheet interface or a local computer terminal.
- The request is automatically stored in the **cloud database** (Google Sheets).
- The Raspberry Pi accesses this data and retrieves the **location code** of the requested book, which contains rack number, shelf level, and row details.

2. Navigation Phase

- The robot starts from its docking station and uses a downward-facing camera to scan floor QR codes for navigation.
- Each QR code provides location data (e.g., aisle and rack number), helping the Raspberry Pi guide movement through the ESP32 motor controller.
- IR and ultrasonic sensors detect obstacles, ensuring safe and accurate autonomous navigation.

3. Rack and Row Scanning

- The robot stops at the correct rack and activates the arm camera to scan QR codes on book spines or shelves.
- It identifies the correct book by matching the scanned code with the requested ID.
- Once confirmed, it prepares for retrieval.

4. Book Retrieval Phase

- The robotic arm moves to the target position and grips the book securely.
- The book is lifted and placed in the carrying tray.
- The OLED screen displays "Book Retrieved Successfully."

5. Return and Delivery Phase

- The robot follows QR markers to return to the librarian's desk.
- Sensors ensure collision-free movement during the return.
- The gripper releases the book into the tray, and the system updates the status as "Delivered."

6. System Reset and Logging

- After completing the operation, the robot resets to **idle mode** at its docking station.
- All operation details, including **book ID, time of retrieval, duration, and user ID**, are logged to the cloud for record-keeping and analytics.

CHAPTER - 6

HARDWARE AND SOFTWARE COMPONENTS

6.1 HARDWARE COMPONENTS:

- Raspberry Pi 4 (main processing unit)
- Raspberry Pi Camera Module
- IR Sensors (for line detection and obstacle avoidance)
- DC Motors and L298N Motor Driver (for locomotion)
- Robotic Arm with Gripper Mechanism (for book handling)
- OLED Display (for system feedback and operation status)
- Power Supply and Jumper Wires (for interconnections and stability)

6.1.1 RASPBERRY PI 4



Figure 6.1.1 : **Raspberry Pi 4**

The **Raspberry Pi 4 Model B** is the central processing unit of the project. It functions as the robot's "brain," handling computational tasks, image processing, and communication between hardware modules. It is a single-board computer that provides high-speed processing, onboard GPIO control, and networking capabilities.

Key Features:

- **Processor:** Quad-core ARM Cortex-A72, 1.5 GHz
- **RAM:** 4GB/8GB LPDDR4
- **Operating System:** Raspberry Pi OS (Linux-based)
- **Connectivity:** Dual-band Wi-Fi, Bluetooth 5.0, USB 3.0, HDMI output
- **Power Supply:** 5V/3A via USB-C

Functions in ILAB:

- The Raspberry Pi runs the core navigation algorithm, using **OpenCV** to detect and decode QR codes placed around the library.
- The Pi Camera continuously captures images, which the Raspberry Pi processes for **object identification**, shelf detection, book label reading, and basic obstacle recognition.
- The Raspberry Pi communicates with the **ESP32 microcontroller** via serial communication (UART). It sends movement commands such as forward, stop, left, or right, while the ESP32 handles low-level motor control and returns sensor feedback.
- It connects to **Google Cloud** to fetch book request data, including book ID, shelf number, and user request details. The Raspberry Pi also updates the cloud database with the robot's current status, task completion, and operational logs.
- An **OLED display** connected to the Raspberry Pi provides real-time information such as current task, navigation status, decoded QR data, obstacle alerts, and system health indicators.

The Raspberry Pi is ideal for this project because it combines **processing power and peripheral control** in a compact, energy-efficient form factor, allowing real-time execution of computer vision tasks . Also, it delivers strong processing power, supports computer vision, and manages multiple peripherals efficiently in a compact, energy-efficient form.

6.1.2 DC MOTORS



Figure 6.1.2: DC MOTORS

DC (Direct Current) motors are used to drive the robot's wheels, enabling forward, backward, and turning movements. These motors convert electrical energy into mechanical rotational motion.

Specifications:

- Operating Voltage: 6V – 12V
- Speed: 150–300 RPM (depending on load)
- Torque: 0.5–2 kg·cm

Functions in ILAB:

- Enable smooth motion across the library floor.
- Work in coordination with the motor driver (L298N) for direction and speed control.
- Respond to signals from the ESP32 microcontroller, which regulates motor activity based on QR navigation data.

The DC motors provide **stable motion** and are cost-efficient, making them ideal for low-weight mobile robotic platforms like ILAB.

6.1.3 RASPBERRY PI CAMERA MODULE



Figure 6.1.3 : Raspberry Pi Camera Module

The Raspberry Pi Camera Module serves as the **primary vision sensor** for the robot. It captures live video frames and transmits them to the Raspberry Pi for image processing.

Specifications:

- Resolution: 8–12 MP (depending on model)
- Frame Rate: Up to 60 FPS
- Interface: CSI (Camera Serial Interface)
- Lens Type: Fixed Focus

Functions in ILAB:

- The **downward-facing camera** detects floor QR codes for navigation.
- The **front-facing camera** mounted on the robotic arm scans shelf and book QR codes for book identification.
- Captures environmental images for visual validation and debugging.

By integrating with **OpenCV**, the camera helps the robot achieve accurate localization and item detection even in dynamic lighting conditions.

6.1.4 L298N MOTOR DRIVER MODULE



Figure 6.1.4 : L298N Motor driver

The **L298N Motor Driver** acts as an interface between the ESP32 controller and the DC motors. It provides sufficient current to drive the motors while protecting the control circuitry from electrical overload.

Key Features:

- Dual H-Bridge configuration
- Supports up to 2A per channel
- Operates between 5V and 35V
- PWM (Pulse Width Modulation) support for speed control

Functions in ILAB:

- Controls the **direction and velocity** of each motor independently.
- Receives signals from the ESP32 microcontroller and translates them into movement commands.
- Supports bidirectional motion (forward/reverse) and turning maneuvers.

The L298N ensures reliable motion control while maintaining low power consumption and precise maneuverability within confined library aisles.

6.1.5 IR SENSOR MODULE



Figure 6.1.5: IR Sensor Module

Infrared (IR) sensors are essential for **line detection, obstacle detection, and positional accuracy**. Each module contains an IR emitter and receiver pair that detects light reflected from surfaces.

Functions in ILAB:

☐ **Line Detection:**

IR sensors detect black and white patterns on the floor by measuring reflected infrared light.

- White surfaces reflect more IR light.
- Black surfaces absorb IR light.

This difference helps the robot accurately follow predefined paths.

☐ **Path Accuracy Verification:**

The IR sensor readings continuously confirm whether the robot is aligned with the correct floor line.

- If the robot deviates, the sensors detect the shift and correct the movement.

☐ **Works With QR Camera for Position Correction:**

The QR camera provides major location checkpoints, while IR sensors fine-tune short-range alignment.

- This combination reduces navigation errors and drift.
- Ensures the robot stays centered on pathways between QR markers.

☐ **Obstacle Detection:**

IR sensors detect nearby objects by sensing changes in reflected IR intensity.

- Helps identify obstacles early.
- Prevents collisions by triggering slow-down or stop commands.

☐ **Real-Time Environmental Awareness:**

Multiple IR sensors placed at the front and sides give the robot a wide detection field.

- Allows smooth turns, corner detection, and safe movement.
- Improves stability in dynamic library environments.

The combination of multiple IR sensors at the front and sides enables real-time **environmental awareness**, ensuring smooth and safe navigation.

6.2 SOFTWARE REQUIREMENTS:

The software setup provides the intelligence behind navigation, control, and cloud connectivity. It includes the following tools and technologies:

1. **Raspberry Pi OS (Operating System)**
2. **Python, OpenCV, TensorFlow Lite / PyTorch Lite**
3. **Google Sheets & Google Cloud Storage**
4. **AppSheet Integration**

6.2.1 Raspberry Pi OS

Raspberry Pi OS, formerly known as Raspbian, is a Linux-based operating system optimized for Raspberry Pi boards.

Functions in ILAB:

- Hosts and executes **Python scripts** that manage camera operations, image processing, and QR decoding.
- Provides a stable environment for communication with peripheral devices via GPIO and I²C interfaces.
- Supports internet connectivity for **cloud synchronization** with Google Sheets.

Its open-source flexibility allows custom driver installation and software updates tailored to the robot's operational needs.

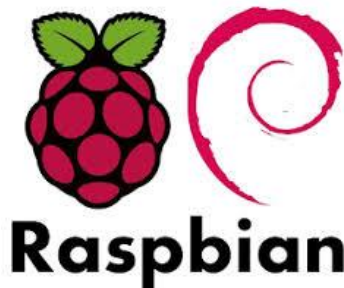


Figure 6.2.1: Raspberry pi

6.2.2 PYTHON, OPENCV, AND TENSORFLOW LITE / PYTORCH LITE

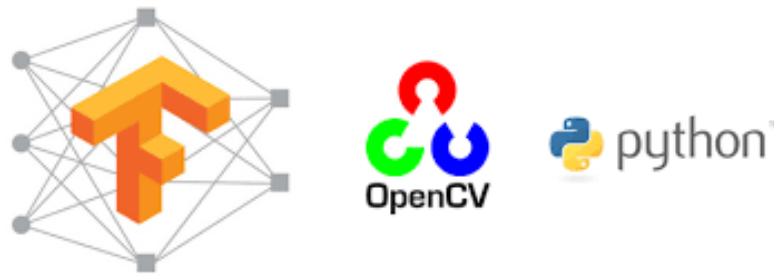


Figure 6.2.2 :python

Python serves as the main programming language due to its simplicity and extensive library ecosystem.

OpenCV (Open Source Computer Vision Library) is used for image acquisition, QR code detection, and visual processing.

6.2.3 GOOGLE CLOUD STORAGE



Figure 6.2.3 : Google cloud

The **Google Cloud** ecosystem provides remote storage, communication, and synchronization for the Intelligent Library Assistance Bot.

Functions in ILAB:

- Stores book records, rack locations, and user requests in Google Sheets.
- Logs retrieval history, timestamps, and performance metrics.
- Enables **remote monitoring** and real-time updates accessible from any device.
- Acts as a bridge between the user interface and the robot through **AppSheet integration**.
- This cloud-based approach ensures **data reliability, scalability, and accessibility**.

CHAPTER - 7

FINAL OUTPUT

7.1 Hardware Setup



Figure 7.1 : Hardware setup

7.2 CODE

```
import cv2
import time
from pyzbar.pyzbar import decode
from pydobot import Dobot

# Camera and scan settings
CAMERA_INDEX = 0
SCAN_DURATION = 3 # seconds to scan before gripping

def scan_book_name(duration=SCAN_DURATION):
    cap = cv2.VideoCapture(CAMERA_INDEX)
    if not cap.isOpened():
        print(f" ERROR: Could not open camera index {CAMERA_INDEX}")
        return None
```

```

print("Scanning book for name...")
start_time = time.time()
detected_name = None

while time.time() - start_time < duration:
    ret, frame = cap.read()
    if not ret:
        continue
    frame=cv2.rotate(frame,cv2.ROTATE_90_CLOCKWISE)
    barcodes = decode(frame)
    for barcode in barcodes:
        barcode_data = barcode.data.decode("utf-8")
        detected_name = barcode_data.strip() # Treat barcode as book name
        print(f" Detected Book Name: {detected_name}")
        break

    cv2.imshow("Book Scan", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        print("Scan aborted by user")
        break
    if detected_name:
        break

cap.release()
cv2.destroyAllWindows()
return detected_name

def pick_book_with_dobot(expected_name, book_positions):
    """
    expected_name: string entered by user
    book_positions: list of dicts [{'x':..., 'y':..., 'z':..., 'r':...}, ...]
    """
    port = "COM3" # Set your Dobot COM port
    device = Dobot(port=port, verbose=True)

    try:
        # Set robot speed and acceleration
        device.speed(velocity=100, acceleration=100)

        for pos in book_positions:

```

```

# Move to target book position
device.move_to(pos['x'], pos['y'], pos['z'], pos.get('r', 0), wait=True)

# Pause 3 seconds before gripping to scan barcode
print("Pausing 3 seconds before gripping for barcode scan...")
time.sleep(3)

# Scan book name
detected_name = scan_book_name()
if detected_name and detected_name.lower() == expected_name.lower():
    print("Book matched! Picking up...")
    device.grip(True) # Close gripper
    time.sleep(3) # Ensure gripper holds book

    # Move to drop-off location
    device.move_to(243.6986846923828, 12.621055603027344, 0.9897842407226562,
48.053733825683594, wait=True)
    device.grip(False) # Release book
    print("Book picked and placed successfully.")
    return True
else:
    print("Book did not match. Moving to next position.")
    device.grip(False) # Ensure gripper is open before next move

print("No matching book found in all positions.")
return False

finally:
    device.close()
if __name__ == "__main__":
    # Get expected book name from user
    user_input = input("Enter the expected book name: ").strip()

    # List of book positions (replace with actual coordinates)
    positions = [
        {'x': 31.91568946838379, 'y': -259.38311767578125, 'z': 7.4087066650390625, 'r': -
37.896263122558594},
        {'x': 35.0, 'y': -260.0, 'z': 7.4, 'r': -37.89},
        {'x': 40.0, 'y': -255.0, 'z': 7.4, 'r': -37.89},
    ]

```



```
pick_book_with_dobot(user_input, positions)
```

7.3 OUTPUT

7.3.1 SOFTWARE OUTPUT

[illegible]

Figure 7.3.1 Software Output

CHAPTER 8

APPLICATIONS AND FUTURE ENHANCEMENT

The *Intelligent Library Assistance Bot* is designed to bring automation and smart technology into library environments. It has wide-ranging applications in educational, public, research, and corporate sectors. The project not only saves time and effort but also enhances the user experience by making library operations smoother, faster, and more reliable. The combination of robotics, IoT, and cloud integration allows this system to work efficiently in different types of organizations. In addition, there are several ways this project can be improved in the future to make it more advanced, user-friendly, and capable of handling larger and more complex environments.

8.1 Applications

1. Educational Institutions

The Intelligent Library Assistance Bot can be very useful in schools, colleges, and universities where thousands of books are used every day.

- The bot can automatically locate and retrieve academic books based on the user's request, saving time for both students and library staff.
- It helps manage large book collections efficiently without needing additional manpower.
- During busy hours, it can reduce waiting time by quickly fetching requested books and updating the records instantly.
- This system makes the library more modern, interactive, and efficient for educational use.

2. Public Libraries

In public libraries, where a large number of people visit daily, this bot can significantly improve accessibility and service quality.

- It can locate and deliver books quickly, helping readers find what they need without manually searching shelves.

- It can also assist *differently-abled* individuals who may find it difficult to move around and search for books physically.
- By automating book retrieval and record keeping, it promotes the concept of a *smart public library* that uses technology to serve the community better.
- The integration of digital records also supports better management and easy monitoring of book transactions.

3. Research Centers and Laboratories

In research environments, quick and accurate access to technical materials is very important.

- The bot can help researchers easily find journals, technical documents, and reports without wasting time.
- Through cloud synchronization, all records are stored and updated in real time, ensuring proper tracking of borrowed and returned materials.
- This system reduces the manual work of maintaining logbooks or databases and improves accuracy in managing research-related resources.

4. Corporate and Private Libraries

Private organizations, companies, and industries often maintain libraries for technical manuals, project documentation, and design references.

- The Intelligent Library Assistance Bot can help employees quickly locate required files or books, thereby saving time and improving productivity.
- It can be used as part of a company's knowledge management system to keep digital and physical records synchronized.
- The bot provides a modern and efficient approach to managing large volumes of technical information in offices and corporate libraries.

8.2 Future Enhancement

Although the current prototype performs effectively and meets its intended goals, there is always room for improvement. By adding more advanced features, the bot can become more intelligent, efficient, and adaptable to complex environments. The following enhancements can be considered for future development:

1. Voice-Assisted Interaction

- Voice control can be introduced to make the system easier to use.
- Users can simply give voice commands such as “Find the Python Programming book” or “Return Data Structures book,” making interaction natural and simple.
- The bot can also provide voice feedback to confirm its actions or inform users about its progress, such as “Book found” or “Moving to shelf B3.”
- This would make the system more interactive and user-friendly.

4. Automated Return System

- The current prototype mainly focuses on book retrieval, but future versions can also include an automatic return system.
- When a user wants to return a book, the bot can detect the request and place the book back in its correct position on the shelf.
- The database will then automatically mark the book as “Returned,” updating the cloud records without manual intervention.
- This enhancement would complete the cycle of borrowing and returning books automatically.

5. AI-Based Shelf Mapping

- Another important improvement could be the use of **SLAM (Simultaneous Localization and Mapping)** technology.
- SLAM allows the bot to create a dynamic map of the library and update it automatically if shelves or layouts change. This helps the bot plan the shortest and safest path to the destination.

CHAPTER 9

9.1 CONCLUSION:

The **Intelligent Library Assistance Bot** aims to modernize traditional library management by integrating **automation, robotics, and smart technologies**. It addresses the challenges of manual book retrieval by automating the process of locating, picking, and placing books, thereby improving efficiency, accuracy, and user convenience. Traditional libraries often require significant time and effort to find and manage books, but this project provides a **cost-effective and intelligent solution** that minimizes human involvement while increasing productivity.

The system combines **mechanical design with intelligent programming** using components like the **Raspberry Pi, ESP32 microcontroller, and QR code-based navigation**. The Raspberry Pi functions as the system's brain, performing image processing and decision-making, while the ESP32 controls motors, sensors, and robotic movements. QR codes placed along the library path enable the robot to navigate autonomously and locate specific shelves efficiently.

The software aspect includes **OpenCV** for QR code detection and **Google Sheets with AppSheet** for cloud integration. This allows real-time updates of the library's digital records whenever books are borrowed or returned. Users and librarians can access this data easily through a computer or mobile device, ensuring transparency and convenience.

The system's **scalability and affordability** make it ideal for educational and institutional use. It can also be upgraded with features like **voice command control, AI-based book identification, or multi-robot coordination** for larger libraries. Academically, it provides hands-on learning opportunities in **robotics, IoT, computer vision, and automation**, helping students understand the collaboration between hardware and software in solving real-world problems.

In conclusion, the Intelligent Library Assistance Bot successfully integrates **automation, image processing, and cloud technologies** to create a reliable and efficient library system. It represents a significant step toward building **smart and intelligent libraries**, enhancing user experience while promoting digital transformation in educational environments.

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